

Elias Hossain

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RESEARCH STATEMENT	I am a Ph.D. researcher at the University of Central Florida, advised by Dr. Niloofar Yousefi. I work on making large language models reliable when human supervision does not scale , across the full alignment pipeline. (1) Verifying preference supervision before it reaches RLHF or DPO, so corrupted feedback never becomes a gradient. (2) Making pretrained transformers report calibrated uncertainty and abstain under a risk budget. (3) Monitoring tool-using agents and intervening on unsafe trajectories rather than unsafe outputs. (4) Steering frozen policies at inference time, without retraining or retrieval. One constraint runs through all of it: the method must work on a model I cannot retrain, since that is the access most deployed systems allow, so the work is inference-time or training-free and evaluated under adversarial input and distribution shift. I stress-test these methods in clinical and biomedical settings, where a confident error is measured in patient harm rather than a benchmark delta.	
RESEARCH IMPACT	• 955 citations, h-index 13, i10-index 13 (Google Scholar, August 2026). • First-author work on LLM safety and alignment at the <i>ICLR 2026 Trustworthy AI Workshop</i>, in <i>IEEE Transactions on Systems, Man, and Cybernetics: Systems</i>, and under review at NeurIPS and ACL/EMNLP venues; an invited poster at the Amazon Trusted AI Symposium. • 12+ first-author peer-reviewed publications in Q1 venues, including <i>IEEE T-SMC</i>, <i>Computers in Biology and Medicine</i>, <i>Briefings in Bioinformatics</i>, and <i>Scientific Reports</i> (Nature Portfolio). • Most-cited lead-author work: a systematic review of NLP in electronic health records (507 citations), a deployed retrieval-augmented university chatbot (63 citations), and multimodal brain-tumor segmentation (63 citations). • 5 public benchmarks (74,590+ records), 14+ open-source repositories, and seven project pages with method figures, result tables, and artifacts. • Peer reviewer for 7 international journals (Nature Portfolio, Elsevier, PLOS, SAGE, Taylor & Francis, Springer) and the ICLR Trustworthy AI Workshop.	
EDUCATION	University of Central Florida , College of Engineering and Computer Science, Orlando, FL, USA Ph.D., Industrial Engineering & Management Systems Aug 2025 – Present • Computer Science concentration. Advisor: Dr. Niloofar Yousefi; Co-advisor: Dr. Ivan Garibay. Complex Adaptive Systems Laboratory. • Dissertation: <i>Self-Verifying Scientific AI: A Verification-Centric Framework for Trustworthy Biomedical Reasoning</i>. Qualifying examination passed Summer 2026; expected graduation Spring 2028. Mississippi State University , Mississippi State, MS, USA M.S., Computer Science (Research), GPA 3.88/4.00 Jan 2024 – Jul 2025 • Thesis: <i>A Multi-Stage AI Framework for Topic Discovery in Scientific Abstracts</i>. Advisors: Dr. Andy D. Perkins, Dr. Shahram Rahimi.	
RESEARCH EXPERIENCE	Graduate Research Assistant , Complex Adaptive Systems Laboratory, UCF Aug 2025 – Present • Lead an independent program on trustworthy and safe LLMs: alignment under corrupted preference supervision, calibrated uncertainty for foundation models, runtime safety for tool-using agents, and inference-time control of frozen policies.	

- Peer-reviewed acceptance at the **ICLR 2026 Trustworthy AI Workshop** and an invited poster at the **Amazon Trusted AI Symposium**. First-author manuscripts actively under review at NeurIPS and the ACL/EMNLP venues; published lead-author work appears in Q1 journals.
- Stress-test these methods where confident errors are costly: clinical decision support and antimicrobial-resistance discovery, via CareBench (process-aware evaluation of clinical agents) and BIOGEN (multi-agent transcriptomic reasoning).

Graduate Research Assistant, Mississippi State University

May 2025 – Jul 2025

- Built a contrastive-learning framework for anomaly detection in large-scale cybersecurity event logs, pairing contrastive sampling of rare malicious events with uncertainty estimation so that precision stays stable under heavy class imbalance. The work produced a peer-reviewed ICICT 2026 paper.
- Co-developed **BarkPlug V.2**, a retrieval-augmented chatbot deployed at MSU, with FAISS retrieval evaluated by RAGAS metrics and the System Usability Scale.

INDUSTRY EXPERIENCE

Software Engineer, REVE Systems, Dhaka, Bangladesh

Dec 2022 – Nov 2023

Deployed speech-recognition models (DeepSpeech, Wav2Vec 2.0) with 20% lower word error rate and 15% faster inference; delivered a voice-enabled extension for Bangladesh government e-governance services.

Senior AI/ML Researcher, Time Research & Innovation, UK

Oct 2020 – Nov 2022

Deep-learning architectures for COVID-19 diagnosis from X-ray and CT imagery; co-developed a telemedicine platform for the UK healthcare system.

PEER-REVIEWED PUBLICATIONS

Full list and live citation counts on [Google Scholar](#).

Trustworthy AI and Large Language Models

Elias Hossain, Maryam Rahimimovassagh, Subash Neupane, Mohammad Jahid Ibna Basher, Ivan Garibay, Niloofer Yousefi. “Fault-Tolerant Preference Alignment via Multi-Agent Verification.” *ICLR 2026 Workshop on Principled Design for Trustworthy AI*. [[OpenReview](#), [project page](#)] Four heterogeneous verifier agents gate preference data under a k-of-n consensus rule; consensus provably suppresses corrupted supervision at an exponential rate, cutting summarization refusals from 0.148 to 0.005 at parity win-rate on Qwen2-7B.

Elias Hossain, Md. Mehedi Hasan Bhuiyan Nipu, Mohammad Sakib Mahmood, Md. Jakir Hossen, M. F. Mridha. “Safe and Scalable Collaboration in Multi-Agent LLM Systems: A Comprehensive Review.” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2026. [[link](#)]

Elias Hossain, Maryam Rahimimovassagh, Niloofer Yousefi. “SAVe V.1: Multi-Agent Safe Alignment Verification for Preference-Based LLM Optimization.” *Trusted AI Symposium*, Amazon JFK27, New York, NY, 2026. (Invited poster.)

Elias Hossain, Umesh Biswas, Charan Gudla, Sai Phani Parsa. “Learning Robust Representations for Malicious Content Detection via Contrastive Sampling and Uncertainty Estimation.” *9th International Conference on Information and Computer Technologies (ICICT)*, USA, 2026. [[arXiv](#)]

Applied Machine Learning and Biomedical AI

Elias Hossain, Rajib Rana, Niall Higgins, Jeffrey Soar, Prabal Datta Barua, Anthony R. Pisani, Kathryn Turner. “Natural Language Processing in Electronic Health Records in relation to health-

care decision-making: A systematic review.” *Computers in Biology and Medicine*, 2023. [\[link\]](#) **507 citations**.

Elias Hossain, Tasfia Nuzhat, Shamsul Masum, Shahram Rahimi, Noorbakhsh Amiri Golilarz. “R-GAT: Graph Neural Networks for Cancer Document Classification.” *Scientific Reports* (Nature Portfolio), 2026. [\[link\]](#)

Elias Hossain, Mehrdad Shoeibi, Ivan Garibay, Niloofar Yousefi. “BIOGEN: evidence-grounded multi-agent reasoning framework for transcriptomic interpretation in antimicrobial resistance.” *Frontiers in Bioinformatics*, vol. 6, 2026. [\[link\]](#), [project page](#)

Elias Hossain, Niloofar Yousefi. “Computational paradigms for antimicrobial resistance prediction: integrating multi-omics, structural modeling, and foundation artificial intelligence systems.” *Briefings in Bioinformatics*, vol. 27, no. 3, 2026. [\[link\]](#)

Subash Neupane, **Elias Hossain**, Jason Keith, Himanshu Tripathi, Farbod Ghiasi, Noorbakhsh Amiri Golilarz, Amin Amirlatifi, Sudip Mittal, Shahram Rahimi. “From questions to insightful answers: Building an informed chatbot for university resources.” *IEEE Frontiers in Education Conference (FIE)*, 2024. [\[link\]](#) **63 citations**.

Mehrdad Shoeibi, **Elias Hossain**, et al. “Surformer v1: Transformer-Based Surface Classification Using Tactile and Vision Features.” *Information*, 2025. [\[link\]](#)

Noorbakhsh Amiri Golilarz, **Elias Hossain**, Shahram Rahimi, Hossein Karimi. “A hybrid approach for pattern recognition and interpretation in age-related false memory.” *Frontiers in Psychology*, 2025. [\[link\]](#)

Additional Peer-Reviewed Work

Seven Scopus-indexed Springer Nature book chapters (LNCS, LNNS, CCIS, LNEE, LNICST), and further refereed journal papers including multimodal brain-tumor segmentation (*Computers, Materials & Continua*, 2022; **63 citations**).

PREPRINTS & MANUSCRIPTS UNDER REVIEW

First-author unless noted.

“UAT-LITE: Inference-Time Uncertainty-Aware Attention for Pretrained Transformers.” With S. R. Dipta, S. Neupane, R. Rana, R. Schwartz-Ziv, I. Garibay, N. Yousefi. [\[arXiv\]](#), [project page](#) *ACL ARR meta-review: borderline conference accept; committed to EMNLP 2026*.

“ChannelGuard: Safe Models Do Not Compose into Safe Multi-Agent Systems.” With M. M. H. Nipu, F. T. J. Faria, T. N. Ornee, M. Sheikh. [\[arXiv\]](#), [project page](#) Training-free information-bottleneck gates on every inter-agent channel; per-trace attribution shows undefended pipelines outsource safety to provider filters.

“NEXUS: Structured Runtime Safety for Tool-Using LLM Agents.” With M. M. H. Nipu, T. N. Ornee, R. Rana, N. Yousefi. [\[arXiv\]](#), [project page](#) Graded intervention policy over allow / block / confirm / revise; 27.3-point gain over rule-only selection at 0.205 ms median latency.

“Right Knowledge, Wrong Answer: Test-Time Steering for Temporal Fact Conflicts in Open-Weight Language Models.” With S. Saha, T. N. Ornee, S. S. Jennifer, U. C. Biswas, S. R. Dipta, R. Rana, N. Yousefi. [\[arXiv\]](#), [project page](#) Benchmark of 8,746 verified Wikidata transitions; activation patching localizes the outdated-fact preference to model-specific upper layers.

“When Policies Cannot Be Retrained.” (PoE-Steer.) [arXiv] Closed-form, post-training steering of a frozen policy through product-of-experts composition, for deployments where retraining or fine-tuning is not an option.

“MedBayes-Lite: A Clinical Uncertainty Governance Layer for Risk-Aware Medical Decision Support.” [arXiv] Bayesian uncertainty quantification turned into a routing policy: low-confidence cases escalate to a clinician under an explicit risk budget rather than being answered.

“Agreement Is Not Evidence: Measuring Correlated Failure in LLM Judge Consensus” (CorrFilter). Large-scale audit of whether LLM-judge banks fail independently, the assumption behind k-of-n consensus filtering for alignment data. Supervised by Prof. Ser-Nam Lim. [project page] *Under review*.

OPEN-SOURCE SOFTWARE & PUBLIC BENCHMARKS

All artifacts at huggingface.co/EliasHossain and github.com/eliashossain001.

Benchmarks. [nanobubbleeval](#) (51,566 records, schema-constrained biomedical extraction); [BioDivergence-Silver-v1.0](#) (11,900 claim pairs); [ptc-benchmark](#) (9,250 temporal-conflict QA); [carebench](#) (clinical agent trajectories); [CancerAbstracts](#) (1,874 abstracts).

Project pages (method figures, result tables, artifacts): [MPV](#), [ChannelGuard](#), [NEXUS](#), [UAT-LITE](#), [TAS](#), [CorrFilter](#), [CareBench](#), and [BIOGEN](#). Research code is released at github.com/eliashossain001, including a 0.6B transformer trained from scratch in PyTorch and domain-adapted clinical language models.

SELECTED TALKS & PRESENTATIONS

Talk, “Fault-Tolerant Preference Alignment via Multi-Agent Verification,” Workshop on Principled Design for Trustworthy AI, ICLR 2026

Invited Poster, “SAVe V.1: Multi-Agent Safe Alignment Verification for Preference-Based LLM Optimization,” Trusted AI Symposium, Amazon JFK27, New York 2026

Invited Talk, artificial intelligence in rural American healthcare, Mississippi Health Innovation Conference, Millsaps College, USA. [video] 2024

Talk, “CITE V.1: Evidence-Grounded, LLM-Agentive Interpretation of RNA-Seq Clusters,” Molecular Machine Learning Conference (MoML), MIT Jameel Clinic, Cambridge, MA 2025

Poster, “BIOGEN: Evidence-Grounded Multi-Agent Reasoning for Transcriptomic Interpretation in AMR,” RECOMB Satellite Workshop on Regulatory & Systems Genomics, Greece 2026

SELECTED COLLABORATIONS

Prof. Ser-Nam Lim (Associate Professor of Computer Science, UCF; formerly Senior Research Scientist Manager at Facebook AI / Meta, 2018–2023). Supervises my work on correlated failure in LLM-judge consensus. **Prof. Björn Schuller** (TU München and Imperial College London; CSO, audEERING): affective computing and computer audition. **Prof. Ravid Shwartz-Ziv** (New York University): information-theoretic deep learning. **Prof. Rajib Rana** (University of Southern Queensland): speech and health AI. **Dr. Pegah Khosravi** (Institute for Artificial Intelligence, UCF): foundation models for healthcare. **Dr. Andrea Rivera-Sepulveda** (Pediatrics, Nemours Children’s Health) and **Dr. Niall Higgins** (Queensland University of Technology): clinical evaluation and validation.

HONORS & AWARDS

- **Graduate Presentation Fellowship**, University of Central Florida, 2025. Selective award funding research presentation at the Molecular Machine Learning Conference, MIT Jameel Clinic.
- **Finalist**, Dr. Pradeep P. Thevannoor Innovation Awards, India, 2019. Selected to the final round through international competition. [news]

- **Earlier awards:** Winner, IEEE SS12 Innovation Challenge & Maker Fair (2018) for the **Kidnap Prevention Mobile App**; Divisional Champion & Global Nominee, NASA Space Apps Challenge (2017).

PROFESSIONAL SERVICE

Journal Peer Reviewer, 2024–Present. *Scientific Reports* (Nature Portfolio), *Computers in Biology and Medicine* (Elsevier), *Computer Methods in Biomechanics and Biomedical Engineering* (Taylor & Francis), *PLOS ONE*, *INQUIRY* (SAGE), *Digital Health* (SAGE), and *International Journal of Computational Intelligence Systems* (Springer).

Workshop Peer Reviewer. ICLR Trustworthy AI Workshop, 2026.

TEACHING

Graduate Teaching Assistant, UCF
STA 3032: Probability and Statistics for Engineers.

Aug 2025 – Present

Graduate Teaching Assistant, Mississippi State University
Five graduate courses spanning software testing, computer forensics, machine learning, and artificial intelligence.

Jan 2024 – May 2025

TECHNICAL SKILLS

LLMs & alignment: RLHF, DPO, SFT; agentic systems; inference-time steering of frozen models; continual pretraining; multimodal modeling. **Trustworthy AI:** uncertainty quantification, calibration, Bayesian inference; LLM evaluation and benchmarking; runtime safety monitoring; adversarial robustness; graph neural networks. **Tooling:** PyTorch, Hugging Face Transformers, scikit-learn, LoRA / QLoRA, multi-GPU training, FAISS, RAG pipelines, Python, R, SQL, Docker, Git, Azure, GCP.